Practice Problem Set -06 (Matrix-II)

Q1. **Verify Caley – Hamilton theorem and express $A^6 - 4A^5 + 8A^4 - 12A^3 + 14A^2$ as a linear polynomial in A, if $A = \begin{bmatrix} 1 & 2 \\ -1 & 3 \end{bmatrix}$.**

Ans: $-4A + 5I$

Q2. **Verify Caley – Hamilton theorem and hence find A^{-1}, *where* $A = \begin{bmatrix} 1 & 2 & 4 \\ 2 & 1 & 2 \\ 4 & 2 & 1 \end{bmatrix}$.**

Ans: $A^{-1} = \frac{1}{3}\begin{bmatrix} -1 & 2 & 0 \\ 2 & -5 & 2 \\ 0 & 2 & -1 \end{bmatrix}$

Q3. **Find the latent root and latent vectors and modal matrix B for the matrix $A = \begin{bmatrix} -2 & 2 & -3 \\ 2 & 1 & -6 \\ -1 & -2 & 0 \end{bmatrix}$, Also find the spectral radius of matrix A.**

Ans: *latent root are* $-3, -3, 5$

$$B = \begin{bmatrix} 3 & 2 & 1 \\ 0 & -1 & 2 \\ 1 & 0 & -1 \end{bmatrix}$$

$$\text{spectral radius} = 5$$

Q4. Verify Caley – Hamilton theorem for matrix

$$A = \begin{bmatrix} 2 & -1 & 1 \\ -1 & 2 & -1 \\ 1 & -1 & 2 \end{bmatrix} \text{ and hence find } A^{-1}.$$

Ans: $A^{-1} = \frac{1}{4}\begin{bmatrix} 3 & 1 & -1 \\ 1 & 3 & 1 \\ -1 & 1 & 3 \end{bmatrix}$

Q5. Find eigen values, eigen vectors and model matrix B for matrix $A = \begin{bmatrix} 3 & 1 & 4 \\ 0 & 2 & 6 \\ 0 & 0 & 5 \end{bmatrix}$.

Ans: $B = \begin{bmatrix} 1 & 1 & 3 \\ -1 & 0 & 2 \\ 0 & 0 & 1 \end{bmatrix}$

Q6. If $A = \begin{bmatrix} 1 & 2 & -3 \\ 0 & 3 & 2 \\ 0 & 0 & -2 \end{bmatrix}$, find the eigen values of $3A^3 + 5A^2 - 6A - 2I$.

Ans: $0, 106, 6$

Q7. If $A = \begin{bmatrix} 1 & 1 & 2 \\ -1 & 2 & 1 \\ 0 & 1 & -1 \end{bmatrix}$, find the eigen values of $A^3 + 5A^2 - 2A - 5I$. Also find the spectral radius of A.

Ans: $Eigen\ values\ are\ 19, 1, -1\ and\ spectral\ radius = 2$